Aria Fusion Standard Operating Procedure & User Guide - System Startup and Shutdown
UC Davis Flow Cytometry Shared Resource

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1. You MUST wear proper PPE to operate the Aria Fusion
    a. Labcoat, safety glasses and gloves.
2. Turn on the Biosafety Cabinet
    a. Turn on blower (press blower icon) - when on green light should be solid
    b. Raise hood sash to proper height.
3. Turn on/restart the Aria Fusion PC.
    a. Green button bottom right hand side of hood
4. Power on compressor
    a. Small June Air unit directly left of hood
5. Wait 5min for startup processes
6. Open Diva software and login
    a. User Name: Labname (ShacklettLab) P/W: PI Last Name (Shacklett)
7. Turn on AriaFusion
8. Allow DIVA to connect to the Aria Fusion.
    a. Check connection status on bottom monitor, bottom right side
9. Once connected typically CST Mismatch dialog will popup.
10. Click “Use CST Settings”.

Note: The lasers on the Aria Fusion require 15-20min of warm up time.

11. While the system is warming up check all liquid levels
    a. Using sterile technique, fill the sheath tank to the weld line with fresh Nerl Saline.
        i. Spray the saline box, top of the sheath tank and your gloves.
        ii. Unplug the sheath level sensor.
        iii. Open the sheath tank and place the lid inside the hood.
        iv. Move the tank over to the saline box and carefully fill the tank to the weld line.
        v. Return the sheath tank to the Aria Fusion and reinstall the lid.
        vi. Plug in the level sensor
    b. Remove the waste tank and fill with 1L of 100% bleach.
        i. Disconnect the level sensor from the waste tank.
        ii. Carefully disconnect both waste fluid lines from the wetcart.
            1. This is untreated waste, take care not to splatter.
        iii. Walk the tank over to the sink
        iv. Remove the waste cap, setting it down red side upNEVER upsidedown
        v. Using the graduated cylinder add 1L of bleach to the tank.
        vi. Replace the waste cap and return the tank to the instrument
        vii. Plug in the level sensor.
        viii. Plug in both fluid lines to the wetcart.
12. Open the Aria Fusion flow cell door (Laser light is blocked so no signals can be viewed!)

13. Reinstall the sample line silastic tubing into the pinch valve.
14. Perform Fluidic Startup:
    a. Cytometer -> Fluidic Startup.
    b. Fluidic startup wizard should popup.
    c. Prepare the sheath tank (step one of wizard):
        i. Disconnect the airline from the ETOH tank and connect it to the sheath tank.
        ii. Remove the top (fluid in line) fluid line from the ETOH filter.
        iii. Remove the top fluid line from the ETOH tank and Plug the top line into the sheath tank.
        iv. Grab the Sheath filter (bottom left of wetcart) and plug in the top fluid line.
        v. Carefully lift up the ETOH filter and remove its bottom line (fluid out line)
        vi. Connect the bottom line to the sheath filter.
        vii. Remove the ETOH filter plastic filter holder (between ETOH and Sheath tanks) and place it in the bottom left of the wet cart.
        viii. Install sheath filter into the plastic filter holder.
    d. Ensure closed loop nozzle is installed in the flow cell.
    e. Wait for the fluidic startup to complete.
    f. Install the correct nozzle for sort: 70um, 85um, 100um or 130um.
    g. Burp the sheath filter.
15. Set the correct Instrument configuration:
    a. Cytometer menu –> View Configuration
    b. Wait for the CST instrument configuration window to popup and load.
    c. Click on the proper configuration for the nozzle size you chose.
        i. 70uM: CST Adaptive Bio 5B-2R-6V
        ii. 85uM:
        iii. 100um: 100um_CST Adaptive Bio 5B-2R-6V_v2
    d. Press “Set Current Configuration”.
    e. Press “OK”

Stream Start and Optimization – setting up the break off

16. Select the proper sort setup. (Sort-> Sort Setup -> 70um, 85um, 100um or 130um.)
    a. Current sort setup displayed at top of Stream Window(e.g 100 micron).
17. Set initial stream settings in the Break Off Window.

Size	Freq	Amp	Atten.	Gap	Breakoff
70					
85					
100	30	30	OFF	10	185
130					

18. Check Break Off Window for any fluid or debris in the camera display..
    a. If necessary, clean the breakoff camera lens and strobe with q-tip .
19. In the Breakoff Window press the “X button” at the top of the window to turn on stream.
    a. Check that there are no bubbles in the flow cell.

20. Open the sort block chamber and check that the stream is flowing smoothly from nozzle to center of aspirator.
    a. If not smooth:
        i. Clean the nozzle by sonication.
        ii. Via microscope check nozzle is not obstructed and O-ring integrity is intact.
        iii. Place a tube of DI water (Use sterile DI if sorting aseptic!) onto the sample station.
        iv. Run a clean flow cell. (Instrument ->Cleaning Modes -> Clean Flowcell)
    b. If the stream is not centered on aspirator adjust the aspirator position with hex wrench.
21. Close the sort block chamber.
22. Close the Flowcell door. *or no lasers will hit flowcell
23. In Diva confirm the stream appears in Breakoff Window.
24. Allow the stream to settle for 15 mins.
25. If stream is unstable or break off is not visible - optimize the nozzle frequency:
    a. Adjust the amplitude so the break off is in the middle of Breakoff Window.
    b. Adjust the frequency to achieve the highest breakoff for a given amplitude, if the break goes off camera (up) adjust amplitude to place it back to middle and continue to optimize frequency. This should not vary much from sort to sort once the stream has settled.

26. Now adjust the amplitude (it should be no less than 10 and no greater than 70 units) until the breakoff is within the first 1/3 of the image and Drop 1 is >100 pixels.
    a. Recommendation is to set it at the bottom of the 2nd full visible drop.
27. QC the stream.
    a. Satellite drops should merge within 4 or 5 drops but no more than 6.
    b. Drop shapes should be consistent day to day (take day 1 image for reference)
    c. If there are issues with stream profile, clean and sonicate nozzle.
28. Now hand enter the Drop 1 value in the entry box next to the current measured Drop 1.
29. Verify the Gap is set correctly.
30. Fine tune the amplitude to achieve the set gap.

System Calibration - Cytometry QC (CST)

31. Open CST. (Cytometer menu –> CST)
32. Allow CST to connect to Aria.
    a. Note: You will be unable to interact with Diva while CST is open, you will get a window "wait" hourglass.
33. Ensure that the proper bead lot is selected from the drop-down menu.
    a. Note: Bead lot is located on the stock CST tube NOT the box.
34. If needed, make a fresh CST sample tube.
    a. 400µL sterile DI water (weld line on FACS tube) + 1 drop CST beads.
35. Load CST tube onto the sample station.
36. Click "Run" in CST software, this can take 5-8mins.
37. When finished click "View Report" from the pop up.
    a. Check primary detectors bright bead robust CV are less than 6% (Ideally less than 4.5%).
    b. Check delta PMT voltage on all detectors is less than 50 (Ideally less than 30).
    c. Check Laser Delay
        i. Red Laser: -35.24 +/- 2
        ii. Violet Laser: 37.17 +/- 2

38. Close CST.
39. Once connected make sure you select “Use CST Settings” from the popup.
40. Check the stream profile once more and repeat stream optimization if needed: Steps 10-15
41. Turn on the “Sweet Spot” Stream monitoring algorithm. (Button at top of Breakoff Window)

Sort Setup – Setting the Drop Delay

42. Prepare Accudrop Experiment - not present.
    a. Close your current experiment.
    b. Create a new experiment. (Experiment menu -> New Experiment)
    c. From the Experiment Template Window select Accudrop Drop Delay.
    d. Press “OK.”
43. Open Accudrop Experiment - if already present
    a. In the Accudrop Experiment expand Specimen_001.
    b. Expand Tube_001.
    c. Open the sort layout by double clicking the “Sort_Layout_001” object.
    d. Select Tube_001. Using the “Tube Pointer”.
44. Align the Accudrop laser to the stream.
    a. Find the Side Stream Camera Window in Diva and bring it to an easy to monitor position.
    b. Locate the metal Accudrop laser adjustment knob on the left side of the sort chamber.
    c. Rotate the knob until you see a bright spot appear near the center of the side stream window.
    d. Rotate back and forth to maximize the intensity of the spot.
45. Setup the side stream for Accudrop
    a. Confirm that the Drop 1 value for the stream matches the entry box in the Breakoff Window. If it does not, type the value to match.
    a. Open the flow cell door.
    b. Open the sort chamber and confirm your voltage plates are dry.
    c. If not then:
        i. Stop the stream. (“X” in Break Off Window)
        ii. Remove and dry the plates. (Use plate remove tool)
        iii. Turn the stream back on and let it settle. (“X” in Break Off Window)
    d. Close the sort chamber.
    e. Close the flow cell door.
    f. Turn on the plate voltage.
        i. 70µm: 4500v – 5000v
        ii. 100um: 2500v
    b. Enable “Test Sort”. (Top of Side Stream Window)
    g. Toggle the Accudrop “Optical Filter” button ON/OFF and note the position of the inner left side stream and center waste stream.
    h. Keep toggling the “Optical Filter” button to check progress while adjusting inner left side stream using the “Side Stream, Adjustment Slider” until it is centered in the left-side Accudrop target box (boxes visible only while filter is in place).
    i. Confirm the center waste stream is within the right Accudrop target box.
        i. If the center stream is not in the right-side box, check that the waste stream is centered on the aspirator (Open sort chamber door). If it is centered, then an Accudrop camera adjustment is needed – this is a service call.
    c. Disable “Test Sort”.
    d. Turn OFF plate voltage.

46. Run the Accudrop procedure
    a. If needed, create an Accudrop bead sample.
        i. 1 drop of Accudrop beads into ~1mL of DI (sterile)
    a. Vortex the Accudrop bead sample.
    b. Load Accudrop bead sample into Aria sample station.
    c. Set the flow rate to “1.0” (Acquisition Dashboard)
    d. Click “Load” (Acquisition Dashboard).
    e. Adjust flow rate to ~1,000-4000 beads/sec. (Flow rate found in Acquisition Dashboard)
    f. In the Sort Layout select “Initial” for precision mode.
    g. Press the “Sort” button in the Sort Layout Window.
    h. Press “Cancel” on the popup.
        ii. We do not want to automatically turn on the voltage plates or open the sort drawer.
    i. Manually turn on the voltage plates. (Top of Side Stream Window)
    j. Manually toggle the optical filter to ON. (Top of Side Stream Window)
    k. You should see percentages next to each box.
        i. If drop delay is close, you will see two spots: 1) Center waste and 2) Left sort stream.
        a. If the Drop Delay is way off you will see one spot in the center waste.

    l. Now make adjustments to the Drop Delay (Top right of Side Stream Window) until you have >95% in the left Accudrop target box.
    m. Change the precision mode from “Initial” to “Fine Tune” in the Sort Layout Window.
        i. Your percentage in the left target box will likely drop.
    n. Now make adjustments to the Drop Delay until you have >95% in the left Accudrop target box. (>98% ideal)
    o. Stop the sort (“Sort” button in Sort layout)
    p. Press “Cancel” in popup - do NOT save a sort report.
    q. Shut off the voltage plates,
    r. Toggle the optical filter OFF.
    s. Unload and remove the Accudrop sample.
    t. Accudrop is finished and your Drop Delay is set!
    u. Prepare a FACSs tube of 3-4mLs of both 20% bleach and 20% Contrad.

New Experiment and Voltage Calibration

1. Create a new experiment or open an existing template. (Experiment Menu -> New Experiment)
2. If using an existing template expand the calibration specimen and select the calibration sample tube, then skip to Area Scaling otherwise proceed to next step. (Use “Tube Pointer”)
3. Create a new specimen. (Syringe Icon in Browser Toolbar)
4. Expand the new specimen to reveal “Tube_001”.
5. Select “Tube_001” (Use “Tube Pointer”, arrow symbol to the left of Tube_001).
6. In the newly visible Parameter Tab in Cytometer Window.
    a. Turn on “W” and “H” for SSC and FSC only.
7. Check and update detector labels in Parameter Tab in Cytometer Window.
    a. Open Experiment layout. (Experiment Menu -> Experiment Layout)
    b. Label channels to be used today.
        i. Just type the names into “Tube_001” section.
8. Close Experiment Layout.

9. Generate the following 4 bi-variable dot plots on the Global Worksheet.
    a. FSC-A X SSC-A
    b. SSC-H X SSC-W
    c. FSC-H X FSC-W
    d. FSC- A X FSC-H – this plot will only be used temporality for Area Scaling
10. Generate the rest of the plots you want to use for sort population identification (You need to have a good idea of what analysis approach you want to take) – NO gates yet!
11. Set Flow Rate in Acquisition Dashboard to "1".
12. Conduct Area Scaling – this is sample type dependent.
    a. In the FSC-H X FSC-A plot, create a triangular gate that goes from the very top right corner to the bottom left corner then to the bottom right corner. (A triangle)
    b. Now load your unstained cell control and run it.
    c. Adjust flow rate to achieve 200-500 events/sec (If at flow rate "1" more than 200-500 is OK)
    d. Adjust the FSC and SSC voltages in Parameter Tab in Cytometer Window to put your population of interest on scale and fully resolved.
        i. Generally FSC: >10k and SSC between 10^3 and 10^5
    e. Referencing the FSC-A X FSC-H plot adjust the FSC Area Scaling value in the Laser Tab in the Cytometer Window) until the populations of interest are on or just slightly to the right of the diagonal line of the triangle gate on the FSC-A X FSC-H plot.
        i. This should scale events into having a slightly greater area relative to height.
13. Load fully stained setup sample.
14. Adjust voltages so all populations are on scale and in linear range for all detectors. ($<10^5$)
    a. Do NOT bring down voltages any lower than necessary.
    b. Going higher is OK but generally not necessary.
    c. FSC and SSC should already be set from the Area Scaling procedure, but OK to adjust.
    d. Once everything is on scale, unload sample – do NOT record any data yet.
15. If your setup requires compensation continue below, otherwise skip to Identify Populations for sorting.

Compensation Setup **(This SOP assumes if you are using compensation beads that you have added a matched negative bead (same lot!) to each single-color bead control.**

16. Create Diva auto compensation framework. (Experiment Menu -> Compensation Setup -> Create Compensation Controls)
17. Select "Include Separate Unstained Control Tube" check box.
18. Keep only the fluorochrome entries that say "Generic", delete the rest.
19. Now delete all fluorochrome entries that you are not using.
20. Press "OK" – this will generate a new compensation specimen in Diva with a tube for each fluorochrome you are using plus an unstained tube.
21. Expand the "Compensation Controls" specimen.
22. Select the Unstained Control. (Use "Tube Pointer", arrow symbol to left of tube icon)
    a. Ensure that you are currently viewing the Normal Worksheet. (Top of Worksheet Window should say "Normal Worksheet")
    b. Note: The Unstained Compensation tube automatically generates a histogram for each color in your experiment. This makes it a great setup template for checking compensation signals across all used detectors.
23. Without recording, load the first single-color control and run it. (with the Unstained Tube selected!)

a. Checking each histogram for each channel ensures that all signals are on scale (<10^5).
b. Check that each control is brightest in its own channel (FITC is brightest in FITC not PE)
    i. If they are not, make slight adjustments (increasing parent channel voltage is always best if possible) keeping in mind the intensity of previously checked full stain sample and single color controls - these adjustments will change signal levels for all samples!

24. Repeat step 23 for all single controls.
25. Now, run each single control and record each to the proper named tube.
    a. While the compensation sample is recording do the following:
        i. Adjust - FSC X SSC gate to encompass target population
        ii. Adjust - the positive gate on the histogram to encompass population
        iii. If this is a bead comp sample -> create a new negative gate in the histogram to encompass the negative population
    b. *Note: If all your compensation controls are beads - do NOT record anything to the unstained.*
    c. *Note: If you have some bead and some cell compensation controls, you will need to record unstained cells to the Unstained Tube.*
26. After all comps are recorded and all gates have been set -> calculate the compensation. (Experiment menu -> Compensation Setup -> Calculate Compensation)
27. In the window click "Link and Save."
28. Compensation is now complete and has been applied to your experiment (*Note: the yellow chain link that appears on the instrument settings*).
29. Return your worksheet view to "Global Worksheets." (Small Icon in upper left of Worksheet Window that looks like a page)

WE ARE HERE

Identify Populations for sorting

30. If it is not already, select Tube_001 (Use "Tube Pointer", arrow symbol to left of Tube_001) in the experimental specimen.
31. Label Tube_001 as needed for the full stained setup sample.
32. Load the full stained setup sample. (or sort sample if you do not have setup sample)
33. Record enough to see all your target sort populations.
34. Now you will run and record all controls. (Unstained, FMOs, experimental controls, other gating controls)
35. Click "Next Tube" in the acquisitions dashboard to create a new tube with all the same settings as the previous (*Note: This only works if the "Tube Pointer" is on the last tube of a specimen).*
36. Label this tube with your next sample name.
37. Load that control and record.
38. Repeat above for each control.
39. Now use your controls to generate and set your sort gates.
40. With the gates set, now move on final Sort Setup.

Final Sort Setup and Sorting

1. If the CST Mismatch Window pops up select "Keep Current Settings."
2. If you like, make a new tube and label it. (Something about sort sample)

3. Select the new tube. (use “Tube Pointer”)
4. Create a Sort Layout. (Sort -> New Sort Layout)
5. Set the Device and Precision to the proper value
    a. 2 Tube, Purity mode (common)

6. Enter your sort target population/s into the proper side stream on the Sort Layout Window.
7. Aiming the Side Streams:
    a. Place a dummy tube into each side stream position you plan to use into the proper sort holder.
    b. Insert the sort holder into the Aria
    c. Open Sash.
    d. Open the sort chamber door
    e. Manually turn on the voltage plates. (Button on top of Side Stream Window) ***(WARNING! The high voltage plates are exposed and can shock you.)***
    f. Enable “Test Sort”. (Top of Side Stream Window)
    g. Look inside the sort chamber and visualize the side stream trajectories.
    h. Make any needed adjustments using the “Side Stream Sliders” in the Side Stream Window to get each side stream aimed slightly off center toward the outer edge of each target tube.
    i. Make a mental note for each side stream and its position on the Side Stream Window Camera view.
    j. Disable “Test Sort”. (Top of Side Stream Window)
    k. Manually turn OFF the voltage plates.(Button on top of Side Stream Window)
    l. Close the sort chamber door.
    m. Close Sash.

8. Replace your dummy tubes with your real sort tubes in the proper position. (Make sure to put some media into the bottom of each sort tube)
9. Load your sort sample into the system.
10. As the sample is running, confirm your sort gates.
11. Once confirmed “Sort” in the Sort Layout Window.
12. Say “Yes” to the popup.
    a. We do want the voltage plates to turn on.
    b. We do want the sort drawer to open. (Listen for it!)

13. Your sort is now running.

While Sorting – monitoring

14. Make a mental note once again for each side stream’s position in the Side Stream Window and make sure they do not move.
15. Make a note of the Gap, Drop 1, and stream profile of the stream in the Break Off Window.
16. Monitor your tube volumes so that nothing overflows.
17. If you need to change tubes:
    a. Stop the sort. (“Sort” button in Sort layout)

b. Press “OK” in popup to save a sort report.
c. Press “Stop Acquire” in the Acquisition Dashboard Window.
d. Write down your current sorted target numbers and label each tube with that value.
e. Replace tubes with fresh tubes.
f. Press “Acquire” in the Acquisition Dashboard Window.
g. Press “Sort” Sort Layout Window.
h. Say “Yes” to the popup
    i. We do want the voltage plates to turn on.
    ii. We do want the sort drawer to open. (Listen for it!)

18. You can record data while you are sorting if you would like to capture events.
19. When finished sorting:
    a. Stop the sort (“Sort” button in Sort layout)
    b. Press “OK” in popup to save a sort report.
    c. Manually turn OFF the voltage plates. (Button on top of Side Stream Window)
        i. Manually close Waste Drawer if needed. (Button on top of Side Stream Window)
        ii. Unload and remove the sample tube.
        iii. Collect your sort collection tubes.

Finished Sorting – Cleaning Up

20. Add another tube to your experiment and name it “Cleaning”.
21. Select the Cleaning tube with the Tube Pointer. (Use “Tube Pointer”)
22. Load and Run 3 tubes of cleaning solution for 5mins each on Flow Rate 11.0 (Acquisition Dashboard)
    a. 10% bleach
    b. 70% ETOH
    c. 100% DI Water
23. Turn off the stream (“X” in Break Off Window)
24. Perform a Fluidic Shutdown. (Cytometer -> Fluidic Shutdown)
    a. Follow all on screen prompts.
        i. Once you have moved the pressure line to the ETOH tank make sure you burp the ETOH filter.
        ii. Use sterile DI water for final cleaning solution at end shutdown.
25. Power off the System.
26. Vent pressure on sheath tank. (very important or fluid will drop pH and become gassy).

Shutdown:

2. Normal 5min 10% bleach and Contrad and DI water
3. Either do fluidics shutdown

a. Add 1L 70% ETOH(100% ETOH in cabinet) to ETOH tank
b. Remove custom filtered sheath line and reconnect it to itself and place into hood
c. Connect blue Aria sheath line to ETOH tank/filter
d. Attach pressure line to ETOH tank
e. Install closed loop nozzle
f. Reinstall 50mL tube holder to side of sort chamber.
g. Place closed loop peek tubing into 50mL tube lid with small hole and place 50mL tube into Styrofoam holder
h. Run fluidic shutdown
    i. Use DI water as final cleaning solution.
i. After shutdown is done using the tube of DI water on sit do 2x clean flow cell (Cytometer Menu -> Cleaning Modes ->Clean Flow Cell) with DI water – <u>make sure the DI water tube does not run dry (refill if needed)</u>
j. Reconnect close loop peek tube to main peek line
k. Remove and empty 50mL tube.
l. Power off system.

4. Or if planning to run in next two days
    a. Install closed loop nozzle
    b. Reinstall 50mL tube holder to side of sort chamber.
    c. Place closed loop peek tubing into 50mL tube lid with small hole and place 50mL tube into Styrofoam holder
    d. Using a full tube of DI water on sit do 3x clean flow cell (Cytometer Menu -> Cleaning Modes ->Clean Flow Cell) with DI water – <u>make sure the DI water tube does not run dry (refill if needed)</u>
    e. Reconnect close loop peek tube to main peek line
    f. Remove and empty 50mL tube.
    g. Power off system.
    h. Remove waste tank and refill sheath for next day.